Commercial Leasing Market Analysis

DataFest 2025

1. Table represents Commercial Leasing Trends in Leading Cities

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.2     v tibble    3.2.1
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.0.4
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(dplyr)
library(tidyr)
library(ggplot2)
library(broom)

# Load data
leases <- read.csv("datasets/Leases.csv")

pre_covid_years <- c(2018, 2019)
post_covid_years <- c(2022, 2023, 2024)

leases_clean_cities <- leases %>%
  filter(!is.na(leasedSF), !is.na(city), !is.na(state), !is.na(overall_rent))

# Top 2 cities per state, by total leasing activity
top_cities <- leases_clean_cities %>%
  group_by(state, city) %>%
  summarise(total_leasing = sum(leasedSF, na.rm = TRUE), .groups = "drop") %>%
  arrange(state, desc(total_leasing)) %>%
  group_by(state) %>%
  slice_head(n = 2)

filtered <- leases_clean_cities %>%
  semi_join(top_cities, by = c("state", "city")) %>%
  mutate(period = case_when(
    year %in% pre_covid_years ~ "pre",
```

```
    year %in% post_covid_years ~ "post",
    TRUE ~ NA_character_)) %>%
  filter(!is.na(period))

# One row per city/state/quarter, so each quarter counts once
# regardless of how many lease transactions happened in it
filtered_quarterly <- filtered %>%
  distinct(state, city, year, quarter, period, overall_rent)

#average rent BEFORE COVID (2018-2019), per city
avg_rent_pre <- filtered_quarterly %>%
  filter(period == "pre") %>%
  group_by(state, city) %>%
  summarise(avg_rent_pre = mean(overall_rent, na.rm = TRUE), .groups = "drop")

#average rent AFTER COVID (2022-2024), per city
avg_rent_post <- filtered_quarterly %>%
  filter(period == "post") %>%
  group_by(state, city) %>%
  summarise(avg_rent_post = mean(overall_rent, na.rm = TRUE), .groups = "drop")

#combine and calculate growth rate
growth_rates <- avg_rent_pre %>%
  left_join(avg_rent_post, by = c("state", "city")) %>%
  mutate(growth_rate_percent = 100 * (avg_rent_post - avg_rent_pre) / avg_rent_pre)

#statistical significance (p-value)
p_values <- filtered_quarterly %>%
  group_by(state, city) %>%
  summarise(
    p_value = tryCatch(
      tidy(t.test(overall_rent[period == "post"], overall_rent[period == "pre"]))$p.value,
      error = function(e) NA),
    .groups = "drop")

#combine growth rate + p-value into one table
rent_stats <- growth_rates %>%
  left_join(p_values, by = c("state", "city")) %>%
  arrange(desc(abs(growth_rate_percent))) %>%
  slice_head(n = 24)

#remove duplicate cities that share the same underlying market
city_market_lookup <- leases_clean_cities %>%
  distinct(state, city, market)

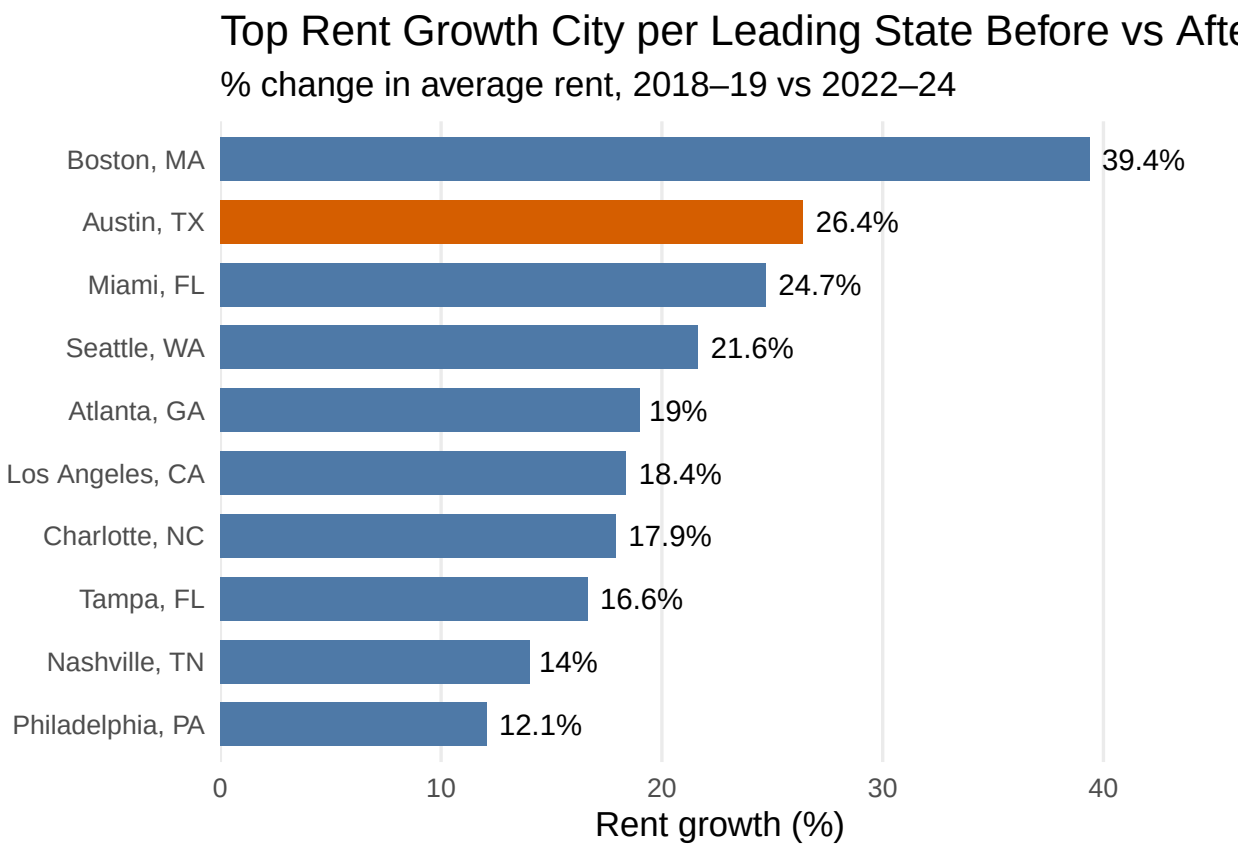
rent_stats_deduped <- rent_stats %>%
  left_join(city_market_lookup, by = c("state", "city")) %>%
  left_join(top_cities %>% select(state, city, total_leasing), by=c("state", "city")) %>%
  group_by(market) %>%
  slice_max(total_leasing, n = 1, with_ties = FALSE) %>%
  ungroup() %>%
  arrange(desc(abs(growth_rate_percent)))
```

```r
print(rent_stats_deduped)
```

```
## # A tibble: 11 x 8
##    state city     avg_rent_pre avg_rent_post growth_rate_percent  p_value market
##    <chr> <chr>           <dbl>         <dbl>                <dbl>    <dbl> <chr>
##  1 MA    Boston           32.7          45.6                 39.4 3.50e-10 Boston
##  2 TX    Austin           36.3          45.9                 26.4 5.22e-10 Austin
##  3 FL    Miami            34.4          42.9                 24.7 3.77e- 9 South~
##  4 WA    Seattle          36.8          44.8                 21.6 1.35e- 6 Seatt~
##  5 GA    Atlanta          26.5          31.6                 19.0 5.33e- 8 Atlan~
##  6 CA    Los Ang~         39.4          46.6                 18.4 1.17e- 5 Los A~
##  7 NC    Charlot~         27.6          32.5                 17.9 2.53e- 8 Charl~
##  8 FL    Tampa            24.7          28.8                 16.6 3.04e- 6 Tampa
##  9 TN    Nashvil~         28.7          32.7                 14.0 3.21e- 6 Nashv~
## 10 PA    Philade~         27.9          31.3                 12.1 9.48e- 7 Phila~
## 11 IL    Schaumb~         23.7          26.6                 11.8 2.84e- 8 Chica~
## # i 1 more variable: total_leasing <dbl>
```

```r
# pick the top 10 for the chart, and highlight Austin
top_growth_cities <- rent_stats_deduped %>%
  arrange(desc(growth_rate_percent)) %>%
  slice_head(n = 10) %>%
  mutate(
    city_label = paste0(city, ", ", state),
    highlight = if_else(city == "Austin", "Austin", "Other"))

# plot
ggplot(top_growth_cities, aes(
    x = reorder(city_label, growth_rate_percent),
    y = growth_rate_percent,
    fill = highlight)) +
  geom_col(width = 0.7) +
  geom_text(aes(label = paste0(round(growth_rate_percent, 1), "%")),
            hjust = -0.15, size = 3.8) +
  coord_flip() +
  scale_fill_manual(values = c("Austin" = "#D55E00", "Other" = "#4E79A7")) +
  scale_y_continuous(limits = c(0, max(top_growth_cities$growth_rate_percent) * 1.15),
                     expand = expansion(mult = c(0, 0))) +
  labs(
    title = "Top Rent Growth City per Leading State Before vs After COVID",
    subtitle = "% change in average rent, 2018-19 vs 2022-24",
    x = NULL, y = "Rent growth (%)"
  ) +
  theme_minimal(base_size = 13) +
  theme(
    legend.position = "none",
    panel.grid.major.y = element_blank(),
    panel.grid.minor = element_blank())
```



2. Pre and Post-COVID Analysis of Leasing Activity (CBD vs Suburban)

```
library(stringr)
library(scales)
```

```
##
## Attaching package: 'scales'

## The following object is masked from 'package:purrr':
##
##     discard

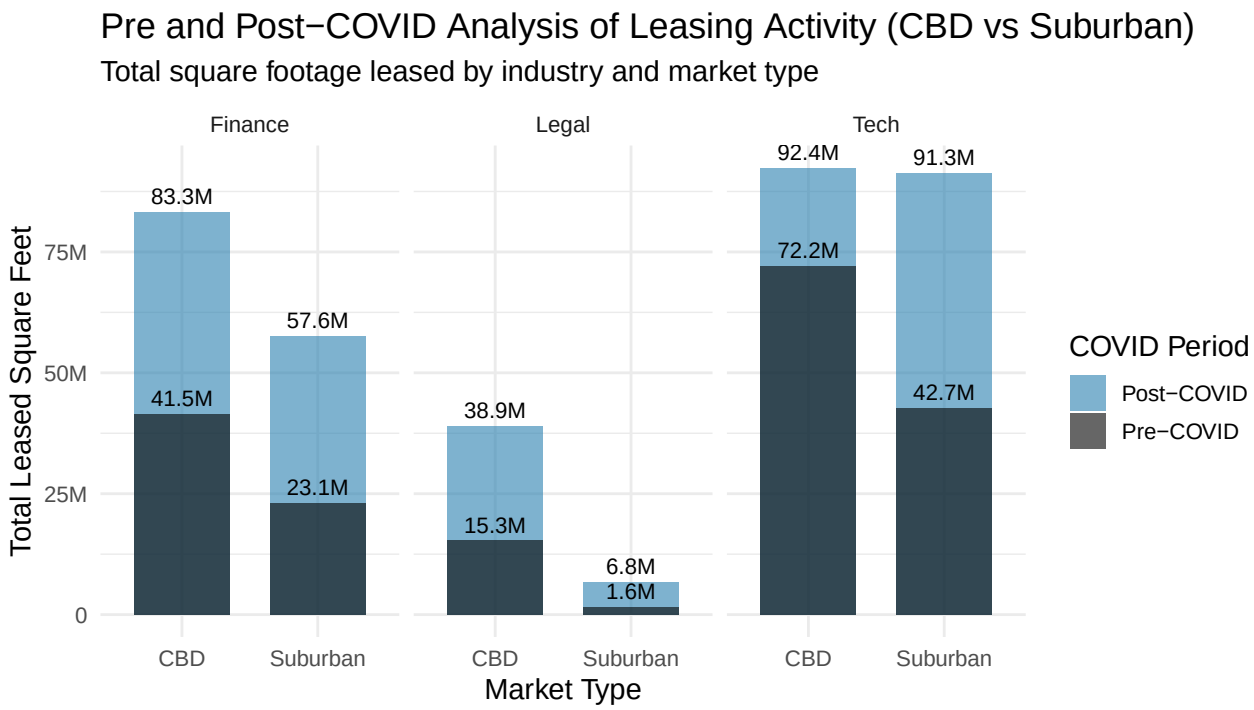
## The following object is masked from 'package:readr':
##
##     col_factor
```

```
leases_clean_industry <- leases %>%
  mutate(
    COVIDPeriod = if_else(year < 2020, "Pre-COVID", "Post-COVID"),
    Industry = case_when(
      str_detect(str_to_lower(internal_industry), "tech|software|it") ~ "Tech",
```

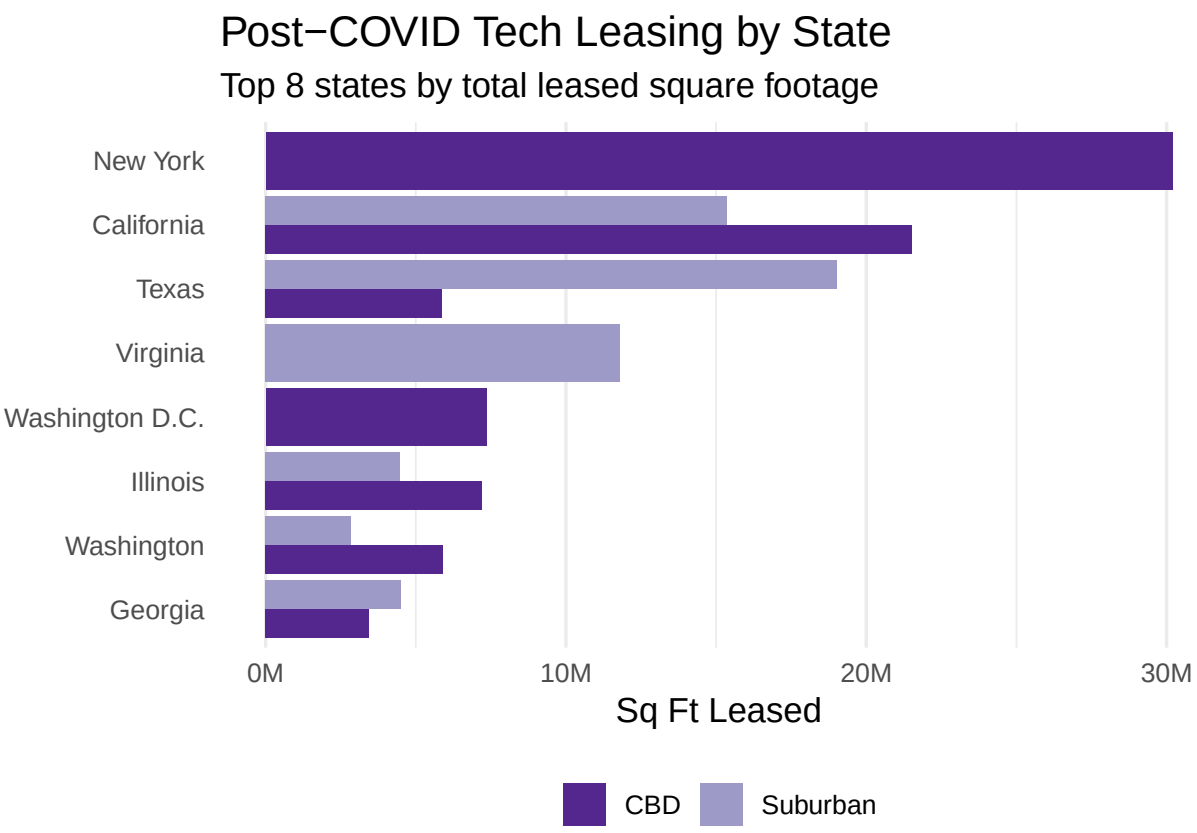
```
      str_detect(str_to_lower(internal_industry),"finance|bank|investment|financial|accounts")~"Finance
      str_detect(str_to_lower(internal_industry), "legal|law") ~ "Legal",
      TRUE ~ NA_character_
    )
  ) %>%
  filter(!is.na(Industry), !is.na(CBD_suburban), !is.na(leasedSF))

suburban_table <- leases_clean_industry %>%
  group_by(CBD_suburban, Industry, COVIDPeriod) %>%
  summarise(TotalLeasedSF = sum(leasedSF, na.rm = TRUE), .groups = "drop") %>%
  arrange(Industry, CBD_suburban, COVIDPeriod)

# Plot
ggplot(suburban_table, aes(x = CBD_suburban, y = TotalLeasedSF, fill = COVIDPeriod)) +
  geom_col(position = "identity", alpha = 0.6, width = 0.7) +
  geom_text(
    aes(label = label_number(scale_cut = cut_short_scale())(TotalLeasedSF)),
    position = position_identity(),
    vjust = -0.5,
    size = 3.5
  ) +
  facet_wrap(~ Industry, scales = "fixed") +
  scale_y_continuous(labels = label_number(scale_cut = cut_short_scale())) +
  scale_fill_manual(values = c("Pre-COVID" = "#000000", "Post-COVID" = "#2880B0")) +
  labs(
    title = "Pre and Post-COVID Analysis of Leasing Activity (CBD vs Suburban)",
    subtitle = "Total square footage leased by industry and market type",
    x = "Market Type",
    y = "Total Leased Square Feet",
    fill = "COVID Period"
  ) +
  theme_minimal(base_size = 13)
```



3. Tech Leasing in CBD/ Suburban Areas (Post-COVID)



4. Growth Rate Trend of Austin pre and post COVID

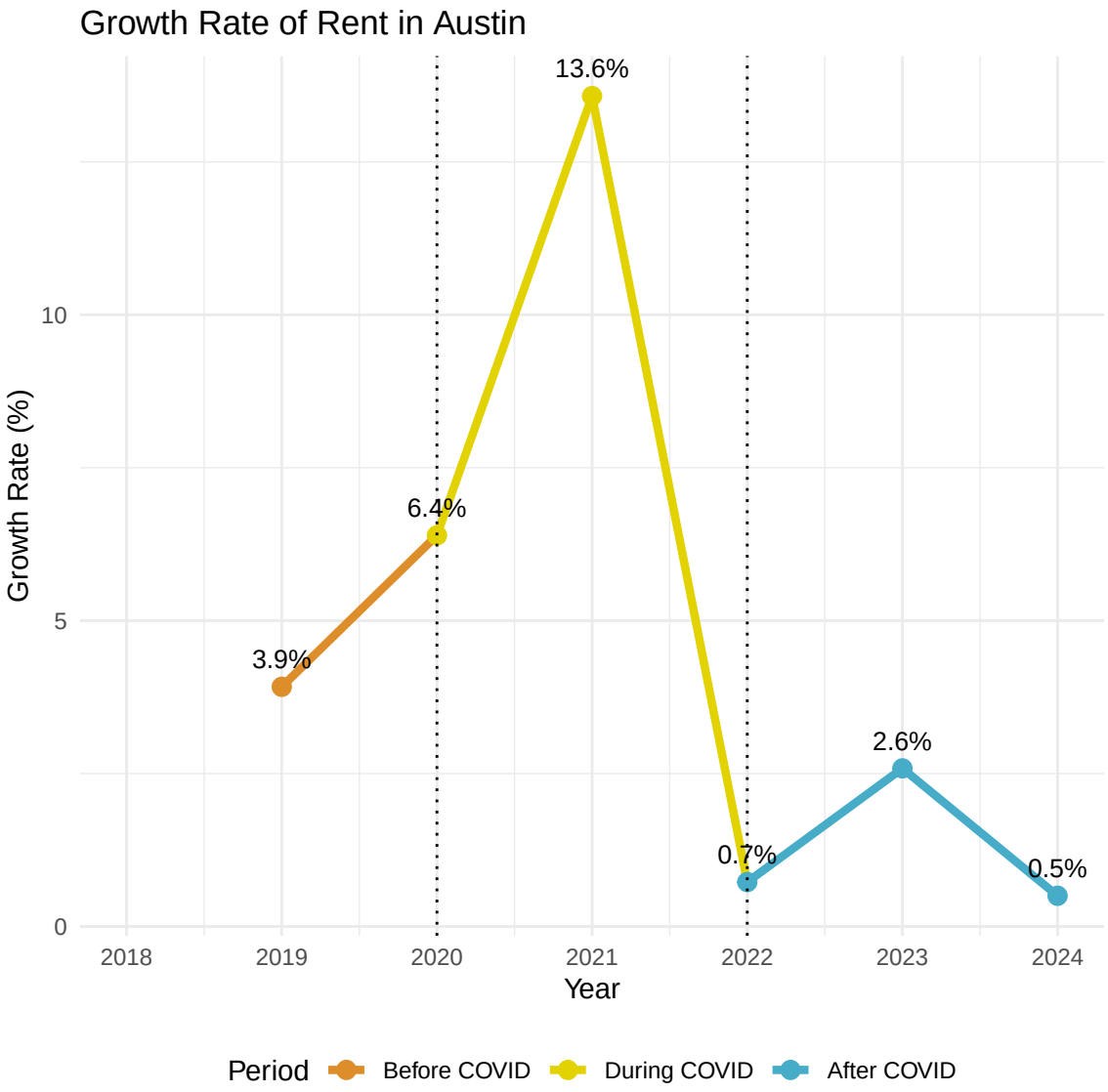
```r
library(wesanderson)

austin_quarterly <- leases %>%
  filter(city == "Austin") %>%
  distinct(year, quarter, overall_rent)

austin_data <- austin_quarterly %>%
  group_by(year) %>%
  summarize(avg_rent = mean(overall_rent, na.rm = TRUE)) %>%
  arrange(year) %>%
  mutate(
    growth_rate = (avg_rent - lag(avg_rent)) / lag(avg_rent) * 100,
    Period = case_when(
      year < 2020 ~ "Before COVID",
      year >= 2020 & year <= 2021 ~ "During COVID",
      year >= 2022 ~ "After COVID"),
    Period = factor(Period, levels = c("Before COVID", "During COVID", "After COVID")))

# Prepare segment lines
austin_segments <- austin_data %>%
  arrange(year) %>%
  mutate(
    x = year,
    xend = lead(year),
    y = growth_rate,
    yend = lead(growth_rate))

# Plot
ggplot(austin_segments) +
  geom_segment(aes(x = x, xend = xend, y = y, yend = yend, color = Period), size = 1.5) +
  geom_point(data = austin_data, aes(x = year, y = growth_rate, color = Period), size = 3) +
  geom_text(data = austin_data,
            aes(x = year, y = growth_rate, label = paste0(round(growth_rate, 1), "%")),
            vjust = -1, size = 3.5) +
  geom_vline(xintercept = 2020, linetype = "dotted", size = 0.5) +
  geom_vline(xintercept = 2022, linetype = "dotted", size = 0.5) +
  scale_color_manual(values = wes_palette("FantasticFox1")[1:3]) +
  labs(title = "Growth Rate of Rent in Austin",
       x = "Year", y = "Growth Rate (%)", color = "Period") +
  scale_x_continuous(breaks = seq(min(austin_data$year), max(austin_data$year), by = 1)) +
  theme_minimal()+
  theme(legend.position = "bottom")
```


Growth Rate of Rent in Austin
13.6%
6.4%
3.9%
0.7%
2.6%
0.5%
10
5
0
Growth Rate (%)
2018
2019
2020
2021
2022
2023
2024
Year
Period
Before COVID
During COVID
After COVID